Matroid Valuation on Independent Sets*

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Abstract

Recently Dress and Wenzel introduced the concept of valuated matroid in terms of a quantitative extension of the basis exchange axiom for matroids. This paper gives two sets of cryptomorphically equivalent axioms of valuated matroids in terms of a function defined on the family of the independent sets of the underlying matroid.

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1 Introduction

Recently Dress-Wenzel [4], [6] introduced the concept of a valuation of a matroid. Let $\mathbf{M} = (V, \mathcal{B})$ be a matroid of rank r defined on a finite set V in terms of the family of bases $\mathcal{B}$ (see, e.g., [18], [19] for matroids), and R be a totally ordered additive group (typically $R = \mathbf{R}$ (reals), $\mathbf{Q}$ (rationals), or $\mathbf{Z}$ (integers)). A valuation of $\mathbf{M} = (V, \mathcal{B})$ is a function $\omega : \mathcal{B} \to R$ which enjoys the following exchange property:

(V1′) For distinct $B, B' \in \mathcal{B}$ and $u \in B - B'$, there exists $v \in B' - B$ such that $B - u + v \in \mathcal{B}$, $B' + u - v \in \mathcal{B}$ and

$$\omega(B) + \omega(B') \leq \omega(B - u + v) + \omega(B' + u - v).$$

A matroid equipped with a valuation is called valuated matroid. Alternatively, we may define a valuated matroid of rank r as a pair (V, ω) of a finite set V and a function $\omega : \mathcal{P}(V, r) \to R \cup \{-\infty\}$, where $\mathcal{P}(V, r)$ denotes the family of subsets of V of cardinality r, such that

(V0) $\omega(B) \neq -\infty$ for some $B \in \mathcal{P}(V, r)$,

(V1) For distinct $B, B' \in \mathcal{P}(V, r)$ and $u \in B - B'$, there exists $v \in B' - B$ such that

$$\omega(B) + \omega(B') \leq \omega(B - u + v) + \omega(B' + u - v).$$

Then $\mathbf{M} = (V, \mathcal{B})$ with

$$\mathcal{B} = \{B \in \mathcal{P}(V, r) \mid \omega(B) \neq -\infty\} \tag{1.1}$$

is a matroid, of which ω (restricted to $\mathcal{B}$) is a valuation. Hence we may regard ω as $\omega : \mathcal{B} \to R$ or interchangeably as $\omega : \mathcal{P}(V, r) \to R \cup \{-\infty\}$ with the agreement (1.1).

A valuation ω can be induced from a weight function $\eta : V \to R$ and $\alpha \in R$ by

$$\omega(B) = \alpha + \sum\{\eta(u) \mid u \in B\} \qquad \text{for } B \in \mathcal{B}. \tag{1.2}$$

Such a valuation is called separable (called "essentially trivial" in [6]). A valuation ω defined by $\omega(B) = 0$ for all $B \in \mathcal{B}$ is called trivial.

The canonical examples of nonseparable valuations, due to [6], are constructed from the determinants of matrices over a valuated field (see Example 4.1). In particular, given a polynomial matrix, the degree of the minors (subdeterminants) defines a variant of valuated matroid, named valuated bimatroid in [9]. See [8], [15] for engineering significances of the degree of the minors.

Nonseparable valuations arise also from graphs as follows [10]. Let $G = (\overline{V}, \overline{A})$ be a directed graph having no self-loops, and S and T be disjoint subsets of the vertex set $\overline{V}$. By L we denote (the arc set of) a Menger-type vertex-disjoint linking from S to T, and by $\partial^+ L$ the set of its initial vertices (in S). Let $\mathcal{L}$ denote the family of maximum linkings from S to T. As is well known, $\mathcal{B} = \{\partial^+ L \mid L \in \mathcal{L}\}$ defines a matroid $\mathbf{M} = (S, \mathcal{B})$. Given a cost function $c : \overline{A} \to \mathbf{Z}$ such that every cycle has a nonnegative cost,

$$\omega(B) = -\min\{\sum_{a \in L} c(a) \mid \partial^+ L = B, L \in \mathcal{L}\} \qquad (B \in \mathcal{B})$$

is a valuation of $\mathbf{M}$. This construction will be investigated further in Example 3.1.

It has turned out that the valuated matroids afford a nice combinatorial framework to which the optimization algorithms for matroids can be generalized. For example, variants of greedy algorithms work for maximizing a matroid valuation [1], [2], [3], [4], [9] (and conversely this property characterizes a matroid valuation [4]). The weighted matroid intersection algorithm can be extended for maximizing the sum of a pair of matroid valuations [10], [11]. Moreover, Frank's weight splitting theorem [7] for the weighted matroid intersection problem can be generalized [10], [12]. See [15] for an engineering application of the latter result.

Besides these results related to combinatorial optimization, not much is known about valuated matroids, as compared with the richness of the theory of matroid. Though the fundamental constructions such as the dual, the restriction and the contraction have been defined in [6] for valuated matroids, extending the corresponding constructions for matroids, other fundamental constructions like "truncation" and "elongation" have not been investigated for valuated matroids.

Matroid theory notably enjoys a large variety of seemingly different but cryptomorphically equivalent axiom systems (see [19] for the terminology of cryptomorphism). For valuated matroids, on the other hand, only a few axioms (or characterizations) are known. Dress and Wenzel have characterized a valuated matroid in terms of the guaranteed success of a kind of greedy algorithm for maximization in [4] and also in terms of circuit functions (and dually, of hyperplane functions) in [5]. The present author has pointed out in [13] that (V1) is equivalent, under (V0), to either of the following apparently weaker exchange properties:

(V2) For distinct $B, B' \in \mathcal{P}(V, r)$, there exist $u \in B - B'$ and $v \in B' - B$ such that

$$\omega(B) + \omega(B') \leq \omega(B - u + v) + \omega(B' + u - v),$$

(V3) For distinct $B, B' \in \mathcal{P}(V, r)$ and $u \in B - B'$, there exist $v \in B' - B$ and $u' \in B - B'$ such that

$$\omega(B) + \omega(B') \leq \omega(B - u + v) + \omega(B' + u' - v).$$

N.B.: After the submission of this paper, a number of new characterizations have been added; in terms of the maximizers [14], in terms of conjugate function [16], and for a function on a generalized (poly)matroid [17].

The objective of this paper is to provide two slightly different sets of axioms of valuated matroids in terms of a function ζ defined effectively on the family $\mathcal{I}$ of the independent sets of the underlying matroid. To be specific, our result (Theorem 3.5) allows us to say that a valuated matroid is a pair (V, ζ) of a finite set V and a function $\zeta : \mathcal{P}(V) \to R \cup \{-\infty\}$, where $\mathcal{P}(V)$ is the family of all the subsets of V, such that

(I0) $\zeta(I) \neq -\infty$ for some $I \in \mathcal{P}(V)$,

(M1) If $I \subseteq J$, then $\zeta(I) \geq \zeta(J)$,

(M2) If $I \subseteq J$, $|I| < |J|$ and $\zeta(J) \neq -\infty$, there exists $v \in V - I$ such that

$$\zeta(I) = \zeta(I + v),$$

($\mathbf{AUG}_1$) If $|I| = |J| - 1$, there exists $v \in J - I$ such that

$$\zeta(I) + \zeta(J) \leq \zeta(I + v) + \zeta(J - v).$$

This is evidently a quantitative extension of the axioms for independent sets of a matroid in the sense that if $\zeta(I) \in \{0, -\infty\}$, the above conditions are easily seen to be equivalent to the independence axioms for $\mathcal{I} = \{I \in \mathcal{P}(V) \mid \zeta(I) \neq -\infty\}$. In fact, (I0) says that $\mathcal{I} \neq \emptyset$, (M1) reduces to $[I \subseteq J \in \mathcal{I} \Rightarrow I \in \mathcal{I}]$, and ($\mathrm{AUG}_1$) to $[I, J \in \mathcal{I},\ |I| = |J| - 1,\ \exists\, v \in J - I\!:\ I + v \in \mathcal{I}]$, while (M2) is implied by the others.

The cryptomorphism relating $\zeta : \mathcal{P}(V) \to R \cup \{-\infty\}$ to $\omega : \mathcal{P}(V, r) \to R \cup \{-\infty\}$ is given by defining ω to be the restriction of ζ to $\mathcal{P}(V, r)$ with $r = \max\{|I| \mid \zeta(I) \neq -\infty\}$. Conversely, ζ is obtained from ω by

$$\zeta(I) = \max\{\omega(B) \mid B \supseteq I\}.$$

It will also be shown (Theorem 2.1) that the restriction of ζ to $\mathcal{P}(V, k)$ with $k \leq r$ yields a valuated matroid of rank k, which construction is naturally regarded as the "truncation" of a valuated matroid.

This paper is organized as follows. In Section 2 we deal with the truncation and, dually, the elongation. Based on this we show in Section 3 the cryptomorphic equivalence of the exchange properties of ω and ζ. It is also pointed out that ζ is a submodular function (Theorem 3.2). Finally in Section 4 we make a supplementary remark on truncation.

2 Truncation and Elongation

Let $\mathbf{M} = (V, \mathcal{B}, \omega)$ be a valuated matroid of rank r, where $\omega : \mathcal{P}(V, r) \to R \cup \{-\infty\}$ and $\mathcal{B} = \{B \in \mathcal{P}(V, r) \mid \omega(B) \neq -\infty\}$ as in (1.1). For k and l with $0 \leq k \leq r \leq l \leq |V|$ we define $\omega_k : \mathcal{P}(V, k) \to R \cup \{-\infty\}$ and $\omega^l : \mathcal{P}(V, l) \to R \cup \{-\infty\}$ by

$$\begin{aligned} \omega_k(I) &= \max\{\omega(B) \mid B \supseteq I, B \in \mathcal{P}(V, r)\} \qquad (I \in \mathcal{P}(V, k)), &(2.1)\\ \omega^l(J) &= \max\{\omega(B) \mid B \subseteq J, B \in \mathcal{P}(V, r)\} \qquad (J \in \mathcal{P}(V, l)), &(2.2)\end{aligned}$$

where the maximum taken over an empty family is understood to be $-\infty$. The effective domains of definitions of ω_k and ω^l, defined respectively by

$$\begin{aligned} \mathcal{B}_k &= \{I \in \mathcal{P}(V, k) \mid \omega_k(I) \neq -\infty\},\\ \mathcal{B}^l &= \{J \in \mathcal{P}(V, l) \mid \omega^l(J) \neq -\infty\},\end{aligned}$$

can be expressed as

$$\begin{aligned} \mathcal{B}_k &= \{I \in \mathcal{P}(V, k) \mid \exists B \in \mathcal{B} : I \subseteq B\},\\ \mathcal{B}^l &= \{J \in \mathcal{P}(V, l) \mid \exists B \in \mathcal{B} : J \supseteq B\}.\end{aligned}$$

Hence $(V, \mathcal{B}_k)$ and $(V, \mathcal{B}^l)$ are the truncation and the elongation of the (nonvaluated) matroid $(V, \mathcal{B})$, respectively.

The following theorem states that ω_k and ω^l are valuations of $(V, \mathcal{B}_k)$ and $(V, \mathcal{B}^l)$ respectively, showing that the concepts of "truncation" and "elongation" can be defined naturally also for valuated matroids (see Section 4 for an extension). We call the valuated matroid $\mathbf{M}_k = (V, \mathcal{B}_k, \omega_k)$ the "truncation of $\mathbf{M} = (V, \mathcal{B}, \omega)$ to rank k" and $\mathbf{M}^l = (V, \mathcal{B}^l, \omega^l)$ the "elongation of $\mathbf{M}$ to rank l". Note that

$$\omega_k = (((\omega_{r-1})_{r-2}) \cdots)_k, \qquad \omega^l = (((\omega^{r+1})^{r+2}) \cdots)^l \qquad (2.3)$$

and recall from Dress-Wenzel [6] that $\mathbf{M}^* = (V, \mathcal{B}^*, \omega^*)$ defined by

$$\mathcal{B}^* = \{B \mid V - B \in \mathcal{B}\}, \quad \omega^*(B) = \omega(V - B)$$

is a valuated matroid of rank $r^* = |V| - r$, called the dual of $\mathbf{M} = (V, \mathcal{B}, \omega)$.

Theorem 2.1 *Let ω_k and ω^l be defined by (2.1) and (2.2), where $0 \leq k \leq r \leq l \leq |V|$.*

(1) $\mathbf{M}_k = (V, \mathcal{B}_k, \omega_k)$ is a valuated matroid (of rank k).

(2) $\mathbf{M}^l = (V, \mathcal{B}^l, \omega^l)$ is a valuated matroid (of rank l).

(3) $(\omega^*)_{r^*-s} = (\omega^{r+s})^*$ for s with $0 \leq s \leq |V| - r$.

(Proof) First we prove (3). For $I \subseteq V$ with $|I| = r^* - s$ we have

$$\begin{aligned}(\omega^*)_{r^*-s}(I) &= \max\{\omega^*(B') \mid B' \supseteq I\} \\ &= \max\{\omega(B) \mid B \subseteq V - I\} = \omega^{r+s}(V - I) = (\omega^{r+s})^*(I).\end{aligned}$$

With this relation the second claim follows from (1).

We now prove (1), for which it suffices to consider the case $k = r - 1$ because of (2.3). Denote ω_{r-1} by ω'. We are to show that for $I, J \in \mathcal{B}_{r-1}$ and $u \in I - J$ there exists $v \in J - I$ such that

$$\omega'(I) + \omega'(J) \le \omega'(I - u + v) + \omega'(J + u - v).$$

Taking $v_* \in V - I$ and $w_* \in V - J$ such that $\omega'(I) = \omega(I + v_*)$ and $\omega'(J) = \omega(J + w_*)$ we rewrite this as

$$\omega(I + v_*) + \omega(J + w_*) \le \omega'(I - u + v) + \omega'(J + u - v). \tag{2.4}$$

Case 1 [$u = w_*$]: If $v_* \in J$ then (2.4) is satisfied with $v = v_* \in J - I$ since $\omega'(I - u + v) + \omega'(J + u - v) = \omega'(I - w_* + v_*) + \omega'(J + w_* - v_*) \ge \omega(I + v_*) + \omega(J + w_*)$. Otherwise ($v_* \notin J$), apply the exchange axiom (V1) to $(I + v_*, J + w_*)$ and $v_* \in (I + v_*) - (J + w_*)$ to see that there exists $v_1 \in (J + w_*) - (I + v_*) = J - I$ such that $\omega(I + v_*) + \omega(J + w_*) \le \omega(I + v_1) + \omega(J + w_* + v_* - v_1) \le \omega'(I - u + v_1) + \omega'(J + u - v_1) =$ RHS of (2.4) with $v = v_1$.

Case 2 [$u \ne w_*$]: Since $u \in (I + v_*) - (J + w_*)$ we see from the exchange axiom (V1) that

$$\omega(I + v_*) + \omega(J + w_*) \le \omega(I + v_* - u + v_2) + \omega(J + w_* + u - v_2) \tag{2.5}$$

for some $v_2 \in (J + w_*) - (I + v_*)$. If $v_2 \ne w_*$, then $v_2 \in J - I$ and

$$\text{RHS of (2.5)} \le \omega'(I - u + v_2) + \omega'(J + u - v_2) = \text{RHS of (2.4) with } v = v_2.$$

Otherwise we have $v_2 = w_*$, and (2.5) reduces to

$$\omega(I + v_*) + \omega(J + w_*) \le \omega(I + v_* - u + w_*) + \omega(J + u). \tag{2.6}$$

We divide into cases according to whether $v_* \in J$ or not.

Case A [$v_* \in J$]: We have $v_* \in J - I$ and

$$\text{RHS of (2.6)} = \omega(I + v_* - u + w_*) + \omega(J + u) \le \omega'(I - u + v_*) + \omega'(J + u - v_*),$$

which shows that (2.4) is satisfied with $v = v_* \in J - I$.

Case B [$v_* \notin J$]: Apply the exchange axiom (V1) to $(I+v_*-u+w_*, J+u)$ and $v_* \in (I+v_*-u+w_*)-(J+u)$ we see that there exists $v_3 \in (J+u)-(I+v_*-u+w_*) = (J-I)+u$ such that

$$\text{RHS of (2.6)} \leq \omega(I-u+w_*+v_3)+\omega(J+u+v_*-v_3). \tag{2.7}$$

If $v_3 \neq u$, this establishes (2.4) with $v=v_3$ since $v_3 \in J-I$. Otherwise ($v_3=u$) we have

$$\text{RHS of (2.7)} = \omega(I+w_*)+\omega(J+v_*) \leq \omega'(I)+\omega'(J). \tag{2.8}$$

From (2.6), (2.7), (2.8) and the obvious relations $\omega(I+w_*) \leq \omega'(I) = \omega(I+v_*)$ and $\omega(J+v_*) \leq \omega'(J) = \omega(J+w_*)$, it follows that all these inequalities are satisfied with equalities. In particular, we have $\omega(I+w_*) = \omega(I+v_*)$ and $\omega(J+v_*) = \omega(J+w_*)$. Applying the exchange axiom (V1) to $(I+v_*, J+v_*)$ and $u \in (I+v_*)-(J+v_*) = I-J$ we finally see that

$$\begin{aligned} \omega(I+v_*)+\omega(J+v_*) &\leq \omega(I+v_*-u+v_4)+\omega(J+v_*+u-v_4) \\ &\leq \omega'(I-u+v_4)+\omega'(J+u-v_4) \end{aligned}$$

for some $v_4 \in (J+v_*)-(I+v_*) = J-I$. This establishes (2.4) with $v=v_4$. $\square$

Remark 2.1 An alternative proof of Theorem 2.1(1) can be found in [14]. This proof is based on the fact that valuated matroids are induced by bipartite graphs just as ordinary (unvaluated) matroids. $\square$

3 Valuations on Independent Sets

3.1 Results

For a function $\omega : \mathcal{P}(V, r) \to R \cup \{-\infty\}$ in general, we define $\overline{\omega} : \mathcal{P}(V) \to R \cup \{-\infty\}$ by

$$\overline{\omega}(I) = \max\{\omega(B) \mid I \subseteq B \in \mathcal{P}(V, r)\}, \tag{3.1}$$

where the maximum taken over an empty family is understood to be $-\infty$. Such construction has been considered first in Dress-Terhalle [2] with the observation that ζ is a "well-layered map" if ω is a matroid valuation (cf. Remark 3.4 below). Recall from Section 2 that the truncation ω_k of a matroid valuation ω is defined as the restriction of $\overline{\omega}$ to $\mathcal{P}(V, k)$. We have shown in Theorem 2.1 that the exchange property (V1) of ω is inherited by ω_k for each k.

In this section we are interested in how the exchange property of ω can be translated into another exchange property of $\overline{\omega}$ as a whole. This amounts to a quantitative extension of the translation of the basis axiom of a matroid into the independence axiom of a matroid.

For a function $\zeta : \mathcal{P}(V) \to R \cup \{-\infty\}$ we denote by $\mathcal{I}$ the effective domain of definition, i.e.,

$$\mathcal{I} = \mathcal{I}(\zeta) = \{I \in \mathcal{P}(V) \mid \zeta(I) \neq -\infty\}.$$

If $\mathcal{I} \neq \emptyset$, let $\mathcal{B} = \mathcal{B}(\zeta)$ denote the family of the maximal elements of $\mathcal{I}$ with respect to set inclusion; we put $\mathcal{B} = \emptyset$ if $\mathcal{I} = \emptyset$. We denote by $\zeta|_k$ the restriction of ζ to $\mathcal{P}(V, k)$, i.e., $\zeta|_k(I) = \zeta(I)$ for $I \in \mathcal{P}(V, k)$. The effective domain of $\zeta|_k$ is equal to

$$\mathcal{B}_k = \{I \in \mathcal{P}(V, k) \mid \zeta(I) \neq -\infty\} = \mathcal{I} \cap \mathcal{P}(V, k).$$

The following condition is natural to impose in our context.

(IND) $\mathcal{I} = \mathcal{I}(\zeta)$ is the family of the independent sets of a matroid.

By definition, (IND) is equivalent to (I0), (I1), (I2) below.

(I0) $\mathcal{I} \neq \emptyset$.

(I1) $I \subseteq J \in \mathcal{I}$ implies $I \in \mathcal{I}$.

(I2) If $I, J \in \mathcal{I}$ and $|I| = |J| - 1$, there exists $v \in J - I$ such that $I + v \in \mathcal{I}$.

If (IND) is the case, $\mathcal{B} = \mathcal{B}(\zeta)$ agrees with the family of the bases. We call the matroid $(V, \mathcal{B})$ the underlying matroid. When we need to specify the rank r of the underlying matroid, we use (IND$_r$) instead of (IND), namely,

(IND$_r$) $\mathcal{I} = \mathcal{I}(\zeta)$ is the family of the independent sets of a matroid of rank r.

We consider the following properties for ζ. As will be shown in Theorem 3.2, these properties are possessed by $\zeta = \overline{\omega}$ derived from a matroid valuation ω by (3.1). Our main interest lies in the relation between (AUG$_*$) and (EXC$_*$).

(M1) If $I \subseteq J$, then $\zeta(I) \geq \zeta(J)$.

(M2) For $I \in \mathcal{I} - \mathcal{B}$ there exists $v \in V - I$ such that $\zeta(I) = \zeta(I + v)$.

(MAX) $\quad \zeta(I) = \max\{\zeta(B) \mid I \subseteq B \in \mathcal{B}\}$,

where it is understood that the maximum taken over an empty family is equal to $-\infty$.

(AUG$_d$) If $|I| = |J| - d$, there exists $v \in J - I$ such that

$$\zeta(I) + \zeta(J) \leq \zeta(I + v) + \zeta(J - v).$$

[We consider (AUG$_d$) for $d \geq 1$ (integer).]

(AUG$_*$) If $|I| < |J|$, there exists $v \in J - I$ such that

$$\zeta(I) + \zeta(J) \leq \zeta(I + v) + \zeta(J - v).$$

(EXC$_k$) For $I, J \in \mathcal{P}(V, k)$ and $u \in I - J$, there exists $v \in J - I$ such that

$$\zeta(I) + \zeta(J) \leq \zeta(I - u + v) + \zeta(J + u - v).$$

[We can talk of (EXC$_k$) for a function defined only on $\mathcal{P}(V, k)$, and this is nothing but the exchange axiom (V1) for a valuated matroid of rank k.]

(EXC$_*$) For $I, J \in \mathcal{P}(V)$ with $|I| = |J|$ and $u \in I - J$, there exists $v \in J - I$ such that

$$\zeta(I) + \zeta(J) \leq \zeta(I - u + v) + \zeta(J + u - v).$$

(SBM) (Submodularity) For $I, J \in \mathcal{P}(V)$,

$$\zeta(I) + \zeta(J) \geq \zeta(I \cup J) + \zeta(I \cap J).$$

First we observe rather obvious implications.

Lemma 3.1

(1) (M1) $\Longrightarrow$ (I1).

(2) (M1), (M2) $\Longleftrightarrow$ (MAX).

(3) (AUG_1) $\Longrightarrow$ (I2).

(4) (I0), (I1), (EXC_*) $\Longrightarrow$ (IND).

(Proof) (1), (2) and (3) are obvious. For (4), note that (EXC_k) implies that $\mathcal{B}_k$, if nonempty, is the basis family of a matroid. □

The following theorem shows that all the above properties are enjoyed by the function $\zeta = \overline{\omega}$ derived from a matroid valuation ω. It is remarked that our main interest lies in the exchange properties (AUG_*) and (EXC_*), and that the submodularity (SBM) is already known [2].

Theorem 3.2 *If $\omega : \mathcal{P}(V, r) \to R \cup \{-\infty\}$ is a matroid valuation of rank r, then $\overline{\omega}$ defined by (3.1) satisfies* (IND_r), (MAX), (M1), (M2), (EXC_*), (SBM), *and* (AUG_*).

(Proof) (IND_r) and (MAX) are due to the definition (3.1). Then (M1) and (M2) follow from Lemma 3.1(2). (EXC_k) has been shown in Theorem 2.1(1) for $k \leq r$, while it is trivial for $k > r$. Then (SBM) is a consequence by Theorem 3.3(2) below. Finally, (AUG_1) can be shown as follows, which implies (AUG_*) by Theorem 3.3(3) below.

To prove (AUG_1) we may assume by Theorem 2.1 that $|I| + 1 = |J| = r$. We may also assume that $\overline{\omega}(I) \neq -\infty$ and $\overline{\omega}(J) = \omega(J) \neq -\infty$. Take $v_* \in V - I$ such that $\overline{\omega}(I) = \omega(I + v_*)$. We are to show that

$$\omega(I + v_*) + \omega(J) \leq \omega(I + v) + \omega(J - v + w)$$

for some $v \in J - I$ and $w \in V - (J - v)$. In case $v_* \in J$, we may choose $v = w = v_*$. Otherwise we have $v_* \in (I + v_*) - J$ and apply (V1) to get

$$\omega(I + v_*) + \omega(J) \leq \omega(I + v) + \omega(J - v + v_*)$$

for some $v \in J - (I + v_*)$. This shows the desired inequality with $w = v_*$. □

The following theorem clarifies the relationship among the above-mentioned properties for $\zeta : \mathcal{P}(V) \to R \cup \{-\infty\}$. We aim in this theorem to investigate the converse of Theorem 3.2 above. See also Remark 3.3.

Theorem 3.3 *For $\zeta : \mathcal{P}(V) \to R \cup \{-\infty\}$ with* (IND)*, the following implications hold.*

(1) (AUG$_1$) $\Longrightarrow$ (EXC$_*$).

(2) (EXC$_*$)*,* (M1)*,* (M2) $\Longrightarrow$ (SBM)*.*

(3) (AUG$_1$)*,* (SBM) $\Longrightarrow$ (AUG$_*$)*.*

(Proof) (1) This will be proven later by Lemma 3.6 and Lemma 3.7.

(2) It suffices to show

$$\zeta(I) + \zeta(I + v + v') \leq \zeta(I + v) + \zeta(I + v') \qquad (v, v' \in V - I, v \neq v', I \subseteq V), \tag{3.2}$$

which is sometimes referred to as "local submodularity". We may assume $I + v + v' \in \mathcal{I}$. By (IND) and (M2) we see that

$$\zeta(I) = \zeta(I + u + u')$$

for some distinct $u, u' \in V - I$. It suffices to consider the case where $\{u, u'\} \cap \{v, v'\} = \emptyset$. Using (EXC$_k$) for $k = |I| + 2$, we may assume

$$\zeta(I + u + u') + \zeta(I + v + v') \leq \zeta(I + u + v) + \zeta(I + u' + v').$$

Therefore, using (M1) we have

$$\begin{aligned} \zeta(I) + \zeta(I + v + v') &= \zeta(I + u + u') + \zeta(I + v + v') \\ &\leq \zeta(I + u + v) + \zeta(I + u' + v') \leq \zeta(I + v) + \zeta(I + v'), \end{aligned}$$

establishing (SBM).

(3) We prove (AUG$_d$) by induction on $d = |J| - |I|$. The case of $d = 1$ is in the assumption. Consider $I, J \in \mathcal{I}$ with $|J| - |I| \geq 2$, assuming (AUG$_d$) for $d \leq |J| - |I| - 1$. By (IND) there exists $v \in J - I$ such that $I' \equiv I + v \in \mathcal{I}$. Noting $|J| - |I'| = |J| - |I| - 1$, we apply (AUG$_d$) with $d = |J| - |I'|$ to (I', J) to obtain

$$-\infty \neq \zeta(I + v) + \zeta(J) \leq \zeta(I + v + v') + \zeta(J - v')$$

for some $v' \in J - (I + v)$. On the other hand, (SBM) implies

$$\zeta(I) + \zeta(I + v + v') \leq \zeta(I + v) + \zeta(I + v').$$

Adding these and noting $\zeta(I + v) \neq -\infty$ and $\zeta(I + v + v') \neq -\infty$, we obtain

$$\zeta(I) + \zeta(J) \leq \zeta(I + v') + \zeta(J - v').$$

$\square$

Theorem 3.3(1) can be reformulated as follows.

Corollary 3.4 *If ζ satisfies* (IND$_r$) *and* (AUG$_1$), *then* $(V, \zeta|_k)$ *with* $k \le r$ *is a valuated matroid of rank* k. $\square$

Example 3.1 Let $G = (\overline{V}, \overline{A})$ be a directed graph having no self-loops, and let $\overline{\mathcal{L}}$ denote the family of Menger-type vertex-disjoint linkings from S to T. We identify $L \in \overline{\mathcal{L}}$ with the set of arcs contained in L, and denote by $\partial^+ L$ the set of its initial vertices (in S). As is well known, $\mathcal{I} = \{\partial^+ L \mid L \in \overline{\mathcal{L}}\}$ forms the family of the independent sets of a matroid on S. Given a cost function $c : \overline{A} \to \mathbf{Z}$ such that every cycle has a nonnegative cost, we define

$$\begin{aligned} c(L) &= \sum\{c(a) \mid a \in L\} \qquad (L \in \overline{\mathcal{L}}), \\ \zeta(I) &= -\min\{c(L) \mid \partial^+ L = I, L \in \overline{\mathcal{L}}\} \qquad (I \in \mathcal{P}(S)), \end{aligned}$$

where the minimum taken over an empty family is understood to be $+\infty$. Then ζ satisfies (AUG$_*$), as well as (IND), and hence Corollary 3.4 applies. To see this, for $I, J \in \mathcal{I}$ with $|I| < |J|$, take $L_I, L_J \in \overline{\mathcal{L}}$ such that $\zeta(I) = -c(L_I)$, $\zeta(J) = -c(L_J)$, $\partial^+ L_I = I$, $\partial^+ L_J = J$. By a standard augmenting-path argument we see that there is an undirected path P ($\subseteq \overline{A}$) such that $P \subseteq L_I \Delta L_J$, $\partial^+ P \in \partial^+ L_J - \partial^+ L_I$, $\partial^- P \in \partial^- L_J - \partial^- L_I$, $L_I \Delta P \in \overline{\mathcal{L}}$, $L_J \Delta P \in \overline{\mathcal{L}}$, where Δ denotes the symmetric difference of two sets, and $\partial^+ P$ and $\partial^- P$ mean the initial vertex and the terminal vertex of P, respectively. For $v = \partial^+ P$ we have $\partial^+(L_I \Delta P) = I + v$, $\partial^+(L_J \Delta P) = J - v$ and

$$\begin{aligned} \zeta(I) + \zeta(J) &= -c(L_I) - c(L_J) \\ &= -c(L_I \Delta P) - c(L_J \Delta P) \\ &\le -\min\{c(L) \mid \partial^+ L = I + v\} - \min\{c(L) \mid \partial^+ L = J - v\} \\ &= \zeta(I + v) + \zeta(J - v). \end{aligned}$$

Note also that ζ does not satisfy (MAX). See Example 4.1 for the linear-algebra counterpart of this construction. $\square$

It should be noted that the mapping $\zeta \mapsto \zeta|_r$ in Corollary 3.4 is not injective. That is, two different ζ satisfying (IND$_r$) and (AUG$_1$) can give rise to an identical valuated matroid $(V, \zeta|_r)$. To obtain a one-to-one correspondence (cryptomorphism) we need to restrict the class of ζ as follows.

For each nonnegative integer r, denote by Ω_r the set of the matroid valuations of rank r defined on V, i.e.,

$$\Omega_r = \{\omega : \mathcal{P}(V, r) \to R \cup \{-\infty\} \mid \omega \text{ satisfies (V0), (V1)}\}$$

and also define

$$\begin{aligned} Z_r^{(1)} &= \{\zeta : \mathcal{P}(V) \to R \cup \{-\infty\} \mid \zeta \text{ satisfies (IND}_r\text{), (M1), (M2), (AUG}_1\text{)}\}, \\ Z_r^{(*)} &= \{\zeta : \mathcal{P}(V) \to R \cup \{-\infty\} \mid \zeta \text{ satisfies (IND}_r\text{), (M1), (M2), (AUG}_*\text{)}\}. \end{aligned}$$

From Theorem 3.3 we see that

$$Z_r^{(1)} = Z_r^{(*)} \tag{3.3}$$

and henceforth we write Z_r for $Z_r^{(1)} = Z_r^{(*)}$. Also recall from Lemma 3.1(2) that the pair of conditions, (M1) and (M2), in the definition of Z_r can be replaced by (MAX).

We are now in the position to present the main result of this paper, saying to the effect that Ω_r and Z_r are equivalent objects. Based on Theorem 3.2 we can define a map $\varphi_r : \Omega_r \to Z_r$ by setting $\varphi_r(\omega) = \overline{\omega}$ with the notation (3.1). Conversely, Theorem 3.3 shows that we can define another map $\psi_r : Z_r \to \Omega_r$ by $\psi_r(\zeta) = \zeta|_r$ (the restriction of ζ to $\mathcal{P}(V, r)$). These two maps, φ_r and ψ_r, are the inverse of each other, which can be seen from

$$\begin{aligned} \psi_r(\varphi_r(\omega)) = \omega \quad &\Longleftrightarrow \quad \max\{\omega(B') \mid B \subseteq B' \in \mathcal{P}(V, r)\} = \omega(B), \\ \varphi_r(\psi_r(\zeta)) = \zeta \quad &\Longleftrightarrow \quad \max\{\zeta(B) \mid I \subseteq B \in \mathcal{P}(V, r)\} = \zeta(I), \end{aligned}$$

where the first assertion is trivially true since B is the only candidate for B' in the maximization, and the second is nothing but (MAX), which is equivalent to (M1) and (M2) by Lemma 3.1(2).

Summarizing the above, we obtain the following theorem.

Theorem 3.5 *For each nonnegative integer r, the axiom* [(V0), (V1)] *is cryptomorphic to the axiom* [(IND$_r$), (M1), (M2), (AUG$_1$)], *as well as to the axiom* [(IND$_r$), (M1), (M2), (AUG$_*$)], *by means of the cryptomorphisms*

$$\varphi_r : \Omega_r \to Z_r, \qquad \psi_r : Z_r \to \Omega_r$$

defined by

$$\varphi_r(\omega) = \overline{\omega}, \qquad \psi_r(\zeta) = \zeta|_r,$$

where

$$\overline{\omega}(I) = \max\{\omega(B) \mid I \subseteq B \in \mathcal{P}(V, r)\}, \qquad \zeta|_k(B) = \zeta(B).$$

□

Remark 3.1 In Theorem 3.3(1), the condition (AUG$_1$) alone (without (IND)) does not imply (EXC$_k$). Consider, for example, $V = \{v_1, v_2, v_3, v_4\}$, $k = 2$, and $\zeta(I) = 0$ if $I = \{v_1, v_2\}$ or $\{v_3, v_4\}$, and $\zeta(I) = -\infty$ otherwise. □

Remark 3.2 (AUG_2) is not implied by (AUG_1). Consider, for example, $V = \{v_1, v_2\}$, $\zeta(\emptyset) = \zeta(v_1) = \zeta(v_2) = 0$ and $\zeta(V) = 1$. $\square$

Remark 3.3 For $\zeta : \mathcal{P}(V) \to R \cup \{-\infty\}$ with (IND), we have (AUG_2) $\Longrightarrow$ (SBM). In fact, for $v, v' \in V - I, I \subseteq V$, we obtain the local submodularity (3.2) from (AUG_2) applied to I and $I + v + v'$. Combining this with Theorem 3.3(3), we see that (AUG_1), (AUG_2) $\iff$ (AUG_*). $\square$

Remark 3.4 Combining Theorem 3.5 and [2, Theorem 3], we see that if $\zeta : \mathcal{P}(V) \to R \cup \{-\infty\}$ has the properties (IND), (M1), (M2) and (AUG_1), it is a well-layered map in the sense of [2]. In particular, it satisfies the following exchange property:

(WLM) If $I \subseteq J$, $|I| \leq |J| - 3$, $\zeta(I) \neq -\infty$, and $u \in J - I$, there exists $v \in J - (I + u)$ such that

$$\zeta(I + u) + \zeta(J - u) \leq \zeta(I + v) + \zeta(J - v).$$

$\square$

Remark 3.5 Theorem 3.3(1), combined with Theorem 3.2, implies Theorem 2.1 as a special case. Note, however, the proof of Theorem 3.2 relies on Theorem 2.1. $\square$

Remark 3.6 If $\zeta_1, \zeta_2 : \mathcal{P}(V) \to R \cup \{-\infty\}$ satisfy (IND) and (AUG_1), the maximum of $f(I) = \zeta_1(I) + \zeta_2(I)$ over $I \in \mathcal{P}(V)$ can be computed in polynomial time. In fact, Corollary 3.4 implies that, for each k, $\max\{f(I) \mid I \in \mathcal{P}(V, k)\}$ can be computed by means of the valuated matroid intersection algorithms of [11]. The maximum of all these maxima gives the answer. $\square$

3.2 Proof of (AUG_1) $\Longrightarrow$ (EXC_*)

For $\zeta : \mathcal{P}(V) \to R \cup \{-\infty\}$ and $p : V \to R$ define $\zeta_p : \mathcal{P}(V) \to R \cup \{-\infty\}$ by

$$\zeta_p(I) = \zeta(I) + \sum\{p(u) \mid u \in I\} \qquad (I \subseteq V), \tag{3.4}$$

and

$$\begin{aligned} \zeta(I, u, v) &= \zeta(I - u + v) - \zeta(I) \qquad (I \in \mathcal{I}, u \in I, v \in V - I), \\ \zeta_p(I, u, v) &= \zeta_p(I - u + v) - \zeta_p(I) \qquad (I \in \mathcal{I}, u \in I, v \in V - I), \end{aligned}$$

where $\zeta(I,u,v) = \zeta_p(I,u,v) = -\infty$ if $I - u + v \notin \mathcal{I}$. If $I, J \in \mathcal{I}$ we have

$$\begin{aligned}
&\zeta(I-u+v) + \zeta(J+u-v) - \zeta(I) - \zeta(J) \\
&= \zeta(I,u,v) + \zeta(J,v,u) \\
&= \zeta_p(I,u,v) + \zeta_p(J,v,u) \qquad (u \in I - J, v \in J - I). \qquad (3.5)
\end{aligned}$$

This shows that (EXC_k) for ζ is equivalent to (EXC_k) for ζ_p. Also, (AUG_d) for ζ is equivalent to (AUG_d) for ζ_p.

Lemma 3.6 *Let $I \in \mathcal{I}$, $I - J = \{u_0, u_1\}$, $J - I = \{v_0, v_1\}$ (with $u_0 \neq u_1$, $v_0 \neq v_1$) and $p : V \to R$. If $\mathcal{I}$ satisfies* (I1) *and ζ satisfies* (AUG_1)*, then*

$$\zeta_p(J) - \zeta_p(I) \leq \max(\pi_{00} + \pi_{11}, \pi_{01} + \pi_{10}),$$

where $\pi_{ij} = \zeta_p(I, u_i, v_j)$ for $i, j = 0, 1$.

(Proof) We may assume $\zeta_p(J) \neq -\infty$. Then by (I1) we have $\zeta_p((I \cap J) + v) \neq -\infty$ for $v \in \{u_0, u_1, v_0, v_1\}$. Define $\xi : \mathcal{P}(\{u_0, u_1, v_0, v_1\}) \to R \cup \{-\infty\}$ by

$$\xi(K) = \zeta_p((I \cap J) \cup K) - \sum_{v \in K} \zeta_p((I \cap J) + v) \qquad (K \subseteq \{u_0, u_1, v_0, v_1\})$$

as well as short-hand notations such as $\xi(u_i) = \xi(\{u_i\})$ and $\xi(u_i, v_j) = \xi(\{u_i, v_j\})$. Note that $\xi(u_i) = \xi(v_j) = 0$ for $i, j = 0, 1$. The desired inequality can be rewritten as

$$\xi(v_0, v_1) - \xi(u_0, u_1) \leq \max(\xi_{00} + \xi_{11}, \xi_{01} + \xi_{10}),$$

where

$$\xi_{ij} = \xi(u_{1-i}, v_j) - \xi(u_0, u_1) \qquad (i, j = 0, 1).$$

We may assume without loss of generality that $\xi_{00} \geq \xi_{ij}$ $(i, j = 0, 1)$.

By (AUG_1) we obtain

$$\begin{aligned}
\xi(v_0, v_1) + \xi(u_i) &\leq \max(\xi(v_0) + \xi(v_1, u_i), \xi(v_1) + \xi(v_0, u_i)) \quad (i = 0, 1), \\
\xi(u_0, u_1) + \xi(v_1) &\leq \max(\xi(u_0) + \xi(u_1, v_1), \xi(u_1) + \xi(u_0, v_1)), \\
\xi(u_0, v_1) + \xi(u_1) &\leq \max(\xi(u_0) + \xi(v_1, u_1), \xi(v_1) + \xi(u_0, u_1)).
\end{aligned}$$

These inequalities are rewritten respectively as follows:

$$\begin{aligned}
\xi(v_0, v_1) - \xi(u_0, u_1) &\leq \max(\xi_{i0}, \xi_{i1}) \qquad (i = 0, 1), \qquad (3.6) \\
0 &\leq \max(\xi_{01}, \xi_{11}), \qquad (3.7) \\
\xi_{11} &\leq \max(\xi_{01}, 0). \qquad (3.8)
\end{aligned}$$

Case 1 [$\xi_{01} \geq 0$]: It follows from $\xi_{00} \geq \xi_{01} \geq 0$ and (3.6) with $i = 1$ that

$$\xi(v_0, v_1) - \xi(u_0, u_1) \leq \max(\xi_{10}, \xi_{11}) \leq \max(\xi_{10} + \xi_{01}, \xi_{11} + \xi_{00}).$$

Case 2 [$\xi_{01} < 0$]: (3.8) shows $\xi_{11} \leq 0$, and then (3.7) implies $\xi_{11} = 0$. Then the right-hand side of (3.6) with $i = 0$ equals $\xi_{00} = \xi_{00} + \xi_{11}$. □

Lemma 3.7 *For* $\zeta : \mathcal{P}(V) \to R \cup \{-\infty\}$ *with* (IND), (AUG$_1$) *implies* (EXC$_k$) *for any* k.

(Proof) Recalling $\mathcal{B}_k = \{I \in \mathcal{P}(V,k) \mid \zeta(I) \neq -\infty\}$ define

$$\begin{aligned}\mathcal{D}_k \quad = \quad & \{(I,J) \mid I, J \in \mathcal{B}_k, \exists u_* \in I - J, \forall v \in J - I : \\ & \qquad \zeta(I) + \zeta(J) > \zeta(I - u_* + v) + \zeta(J + u_* - v)\},\end{aligned}$$

which denotes the set of pairs (I, J) for which the exchangeability (EXC$_k$) fails. We are to show $\mathcal{D}_k = \emptyset$.

Suppose to the contrary that $\mathcal{D}_k \neq \emptyset$, and take $(I, J) \in \mathcal{D}_k$ such that $|J - I|$ is minimum and let $u_* \in I - J$ be as in the definition of $\mathcal{D}_k$. Define $p : V \to R$ by

$$p(v) = \begin{cases} -\zeta(I, u_*, v) & (v \in J - I, I - u_* + v \in \mathcal{B}_k) \\ \zeta(J, v, u_*) + \varepsilon & (v \in J - I, I - u_* + v \notin \mathcal{B}_k, J + u_* - v \in \mathcal{B}_k) \\ 0 & (\text{otherwise}) \end{cases}$$

with some $\varepsilon > 0$ and consider ζ_p defined in (3.4).

Claim 1:

$$\zeta_p(I, u_*, v) \;=\; 0 \qquad \text{if } v \in J - I, I - u_* + v \in \mathcal{B}_k, \tag{3.9}$$

$$\zeta_p(J, v, u_*) \;<\; 0 \qquad \text{for } v \in J - I. \tag{3.10}$$

The equality (3.9) is immediate from the definitions. The inequality (3.10) can be shown as follows. If $I - u_* + v \in \mathcal{B}_k$, we have $\zeta_p(I, u_*, v) = 0$ by (3.9) and

$$\zeta_p(I, u_*, v) + \zeta_p(J, v, u_*) = \zeta(I, u_*, v) + \zeta(J, v, u_*) < 0$$

by (3.5) and the definition of u_*. Otherwise we have $\zeta_p(J, v, u_*) = -\varepsilon$ or $-\infty$ according to whether $J + u_* - v \in \mathcal{B}_k$ or not.

Claim 2: There exist $u_0 \in I - J$ and $v_0 \in J - I$ such that $u_0 \neq u_*$ and $J + u_0 - v_0 \in \mathcal{B}_k$.

In fact, (I1) and (AUG_1) imply

$$\exists v_0 \in J - I: \qquad -\infty \neq \zeta(I - u_*) + \zeta(J) \leq \zeta(I - u_* + v_0) + \zeta(J - v_0),$$
$$\exists u_0 \in I - J: \qquad -\infty \neq \zeta(I) + \zeta(J - v_0) \leq \zeta(I - u_0) + \zeta(J + u_0 - v_0).$$

Adding these inequalities we obtain

$$\zeta(I) + \zeta(J) + \zeta(I - u_*) \leq \zeta(I - u_* + v_0) + \zeta(J + u_0 - v_0) + \zeta(I - u_0),$$

from which it follows that $u_0 \neq u_*$ by the definition of u_* and $\zeta(I - u_*) \neq -\infty$. We also have $J + u_0 - v_0 \in \mathcal{B}_k$ since $\zeta(J + u_0 - v_0) \neq -\infty$.

In addition to the conditions imposed in Claim 2 we can further assume

$$\zeta_p(J, v_0, u_0) \geq \zeta_p(J, v, u_0) \qquad (v \in J - I) \tag{3.11}$$

by choosing v_0 appropriately. Put $J' = J + u_0 - v_0$.

Claim 3: $(I, J') \in \mathcal{D}_k$.

To prove this it suffices to show

$$\zeta_p(I, u_*, v) + \zeta_p(J', v, u_*) < 0 \qquad (v \in J' - I).$$

We may restrict ourselves to v with $I - u_* + v \in \mathcal{B}_k$, since otherwise the first term $\zeta_p(I, u_*, v)$ is equal to $-\infty$. For such v the first term is equal to zero by (3.9). For the second term it follows from Lemma 3.6, (3.10) and (3.11) that

$$\begin{aligned}
\zeta_p(J', v, u_*) &= \zeta_p(J + \{u_0, u_*\} - \{v_0, v\}) - \zeta_p(J + u_0 - v_0) \\
&\leq \max\left[\zeta_p(J, v_0, u_0) + \zeta_p(J, v, u_*), \zeta_p(J, v, u_0) + \zeta_p(J, v_0, u_*)\right] \\
&\quad\ -\zeta_p(J, v_0, u_0) \\
&< \max\left[\zeta_p(J, v_0, u_0), \zeta_p(J, v, u_0)\right] - \zeta_p(J, v_0, u_0) \\
&= 0.
\end{aligned}$$

Since $|J' - I| = |J - I| - 1$, Claim 3 contradicts our choice of $(I, J) \in \mathcal{D}_k$. Therefore we conclude $\mathcal{D}_k = \emptyset$. $\square$

4 Concluding Remark on Truncation

Let $\mathbf{M} = (V, \mathcal{B}, \omega)$ be a valuated matroid of rank r, where $\omega : \mathcal{P}(V, r) \to R \cup \{-\infty\}$ and $\mathcal{B} = \{B \in \mathcal{P}(V, r) \mid \omega(B) \neq -\infty\}$ as in (1.1). The truncation defined in Section 2 reflects the fact that $I \subseteq V$ is independent in matroid $(V, \mathcal{B})$ if and only if $I \subseteq B$ for some $B \in \mathcal{B}$. As is well known, another characterization of independence can be made with reference to a fixed spanning set, say S_0. Namely, $I \subseteq V$ is independent in $(V, \mathcal{B})$ if and only if $I \cup J \in \mathcal{B}$ for some $J \subseteq S_0$. This leads us to define $\omega_{k,S_0} : \mathcal{P}(V, k) \to R \cup \{-\infty\}$ (where $0 \leq k \leq r$) by

$$\omega_{k,S_0}(I) = \max\{\omega(B) \mid I \cup S_0 \supseteq B \supseteq I, B \in \mathcal{P}(V, r)\} \qquad (I \in \mathcal{P}(V, k)). \quad (4.1)$$

This function has been considered in Dress-Terhalle [2] in connection to a version of greedy algorithm. Dually, for an independent set I_0, we define $\omega^{l,I_0} : \mathcal{P}(V, l) \to R \cup \{-\infty\}$ (where $r \leq l \leq |V|$) by

$$\omega^{l,I_0}(J) = \max\{\omega(B) \mid J \cap I_0 \subseteq B \subseteq J, B \in \mathcal{P}(V, r)\} \qquad (J \in \mathcal{P}(V, l)). \quad (4.2)$$

The following theorem, a slight extension (actually a corollary) of Theorem 2.1, reveals that these constructions also yield valuated matroids.

Theorem 4.1 *Let ω_{k,S_0} and ω^{l,I_0} be defined by (4.1) and (4.2) for a spanning set S_0 and an independent set I_0, where $0 \leq k \leq r \leq l \leq |V|$.*

(1) $\mathbf{M}_{k,S_0} = (V, \mathcal{B}_k, \omega_{k,S_0})$ is a valuated matroid (of rank k).

(2) $\mathbf{M}^{l,I_0} = (V, \mathcal{B}^l, \omega^{l,I_0})$ is a valuated matroid (of rank l).

(3) $(\omega^*)_{r^*-s,I_0^*} = (\omega^{r+s,I_0})^*$ for s with $0 \leq s \leq |V| - r$, where $I_0^* = V - I_0$.

(Proof) (1) Define $p : V \to R$ by

$$p(u) = \begin{cases} \alpha & (u \in S_0) \\ 0 & (u \in V - S_0) \end{cases}$$

where α is a sufficiently large number, and consider $\omega_p : \mathcal{P}(V, r) \to R \cup \{-\infty\}$ defined by

$$\omega_p(B) = \omega(B) + p(B) = \omega(B) + \sum\{p(u) \mid u \in B\} \qquad (B \in \mathcal{P}(V, r)),$$

which is also a valuation of $(V, \mathcal{B})$. Then we have

$$\omega_{k,S_0}(I) = (\omega_p)_k(I) - p(I) - \alpha(r - k), \quad (4.3)$$

where $p(I) = \sum\{p(u) \mid u \in I\}$ and $(\omega_p)_k$ denotes the truncation of ω_p to rank k defined as (2.1). This expression shows that ω_{k,S_0} is a matroid valuation, since $(\omega_p)_k$ is a matroid valuation by Theorem 2.1.

(2) This follows from (1) and (3) below.

(3) For $I \subseteq V$ with $|I| = r^* - s$ we have

$$\begin{aligned}
(\omega^*)_{r^*-s,I_0^*}(I) &= \max\{\omega^*(B') \mid I \cup I_0^* \supseteq B' \supseteq I\} \\
&= \max\{\omega(B) \mid (V - I) \cap I_0 \subseteq B \subseteq V - I\} \\
&= \omega^{r+s,I_0}(V - I) = (\omega^{r+s,I_0})^*(I).
\end{aligned}$$

□

Let us call $\mathbf{M}_{k,S_0} = (V, \mathcal{B}_k, \omega_{k,S_0})$ the truncation relative to a spanning set S_0, and $\mathbf{M}^{l,I_0} = (V, \mathcal{B}^l, \omega^{l,I_0})$ the elongation relative to an independent set I_0. Note that

$$\omega_k(I) \geq \omega_{k,S_0}(I) \quad (I \in \mathcal{B}_k), \qquad \omega^l(J) \geq \omega^{l,I_0}(J) \quad (J \in \mathcal{B}^l).$$

The following theorem is a corollary to Theorem 3.2. It is remarked that the submodularity (SBM) is already shown in [2].

Theorem 4.2 *If $\omega : \mathcal{P}(V, r) \to R \cup \{-\infty\}$ is a matroid valuation of rank r and $\omega(B_0) \neq -\infty$ for some $B_0 \subseteq S_0$, then $\overline{\omega}_{S_0} : \mathcal{P}(V) \to R \cup \{-\infty\}$ defined by*

$$\overline{\omega}_{S_0}(I) = \max\{\omega(B) \mid I \cup S_0 \supseteq B \supseteq I, B \in \mathcal{P}(V, r)\} \qquad (I \in \mathcal{P}(V))$$

satisfies (IND$_r$), (EXC$_*$), (SBM), *and* (AUG$_*$).

(Proof) From (4.3) we have $\overline{\omega}_{S_0}(I) = \overline{\omega_p}(I) - p(I) - \alpha(r - |I|)$. The assertion follows from Theorem 3.2. □

Example 4.1 Let $A(x)$ be an $m \times n$ matrix with each entry being a polynomial (or rational function) in a variable x, and let R and C denote the row-set and the column-set of $A(x)$. Define $\zeta : \mathcal{P}(C) \to \mathbf{Z} \cup \{-\infty\}$ by

$$\zeta(J) = \max\{\deg_x \det A[I, J] \mid I \subseteq R, |I| = |J|\} \qquad (J \in \mathcal{P}(C)),$$

where $A[I, J]$ is the submatrix of $A(x)$ with row indices in I and column indices in J and $\deg_x 0 = -\infty$. Then ζ satisfies (AUG$_*$), as well as (IND), and hence $(C, \zeta|_k)$ is a valuated matroid of rank k, provided k is less than or equal to the rank of $A(x)$. This fact can be understood from Theorem 4.2 as follows, though a straightforward linear algebraic proof is also possible. Consider an $m \times (m + n)$ matrix $\tilde{A}(x) =$

$[I_m \mid A(x)]$ and let B_0 denote the set of column indices corresponding to the $m \times m$ identity matrix I_m. The matrix $\tilde{A}(x)$ defines (cf. [6]) a valuated matroid $(B_0 \cup C, \omega)$ with

$$\omega(J) = \deg_x \det \tilde{A}[R, J] \qquad (J \in \mathcal{P}(B_0 \cup C, m)).$$

It is easy to see that ζ is the restriction of $\overline{\omega}_{B_0}$ to C, and therefore $(C, \zeta|_k)$ agrees with the restriction to C of the truncation of $(B_0 \cup C, \omega)$ to rank k relative to B_0. $\square$

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References

[1] A. W. M. Dress and W. Terhalle, Well-layered maps — A class of greedily optimizable set functions, *Applied Mathematics Letters* **8** (1995), 77–80.

[2] A. W. M. Dress and W. Terhalle, Well-layered maps and the maximum-degree $k \times k$-subdeterminant of a matrix of rational functions, *Applied Mathematics Letters* **8** (1995), 19–23.

[3] A. W. M. Dress and W. Terhalle, Rewarding maps — On greedy optimization of set functions, *Advances in Applied Mathematics* **16** (1995), 464–483.

[4] A. W. M. Dress and W. Wenzel, Valuated matroid: A new look at the greedy algorithm, *Applied Mathematics Letters* **3** (1990), 33–35.

[5] A. W. M. Dress and W. Wenzel, Perfect matroids, *Advances in Mathematics* **91** (1992), 158–208.

[6] A. W. M. Dress and W. Wenzel, Valuated matroids, *Advances in Mathematics* **93** (1992), 214–250.

[7] A. Frank, A weighted matroid intersection algorithm, *Journal of Algorithms* **2** (1981), 328–336.

[8] K. Murota, Combinatorial relaxation algorithm for the maximum degree of subdeterminants: Computing Smith-McMillan form at infinity and structural indices in Kronecker form, *Applicable Algebra in Engineering, Communication and Computing* **6** (1995), 251–273.

[9] K. Murota, Finding optimal minors of valuated bimatroids, *Applied Mathematics Letters* **8** (1995), 37–42.

[10] K. Murota, Valuated matroid intersection, I: optimality criteria, *SIAM Journal on Discrete Mathematics* **9** (1996), to appear.

[11] K. Murota, Valuated matroid intersection, II: algorithms, *SIAM Journal on Discrete Mathematics* **9** (1996), to appear.

[12] K. Murota, Fenchel-type duality for matroid valuations, Report No. 95839-OR, Forschungsinstitut für Diskrete Mathematik, Universität Bonn, 1995.

[13] K. Murota, On exchange axioms for valuated matroids and valuated delta-matroids, *Combinatorica* **16** (1996), to appear.

[14] K. Murota, Convexity and Steinitz's exchange property, *Advances in Mathematics*, to appear. Extended abstract in: Integer Programming and Combinatorial Optimization (W. H. Cunningham, S. T. McCormick, and M. Queyranne, eds.), Proceedings of 5th International IPCO Conference, Vancouver, British Columbia, Canada, June 3-5, 1996. Lecture Notes in Computer Science **1084**, Springer-Verlag, 1996, 260–274.

[15] K. Murota, Structural approach in systems analysis by mixed matrices – An exposition for index of DAE, *ICIAM 95 (Proceedings of the Third International Congress on Industrial and Applied Mathematics held in Hamburg, Germany, July 3-7, 1995)*, K. Kirchgässner, O. Mahrenholtz, and R. Mennicken, eds., Mathematical Research **87** (1996), Akademie Verlag, 257–279.

[16] K. Murota, Discrete convex analysis, RIMS preprint No. 1065, Kyoto University (1996).

[17] K. Murota and A. Shioura, M-convex function on generalized polymatroid, RIMS preprint, 1996.

[18] D. J. A. Welsh, "Matroid Theory,"Academic Press, London, 1976.

[19] N. White (ed.), "Theory of Matroids," Cambridge University Press, London, 1986.